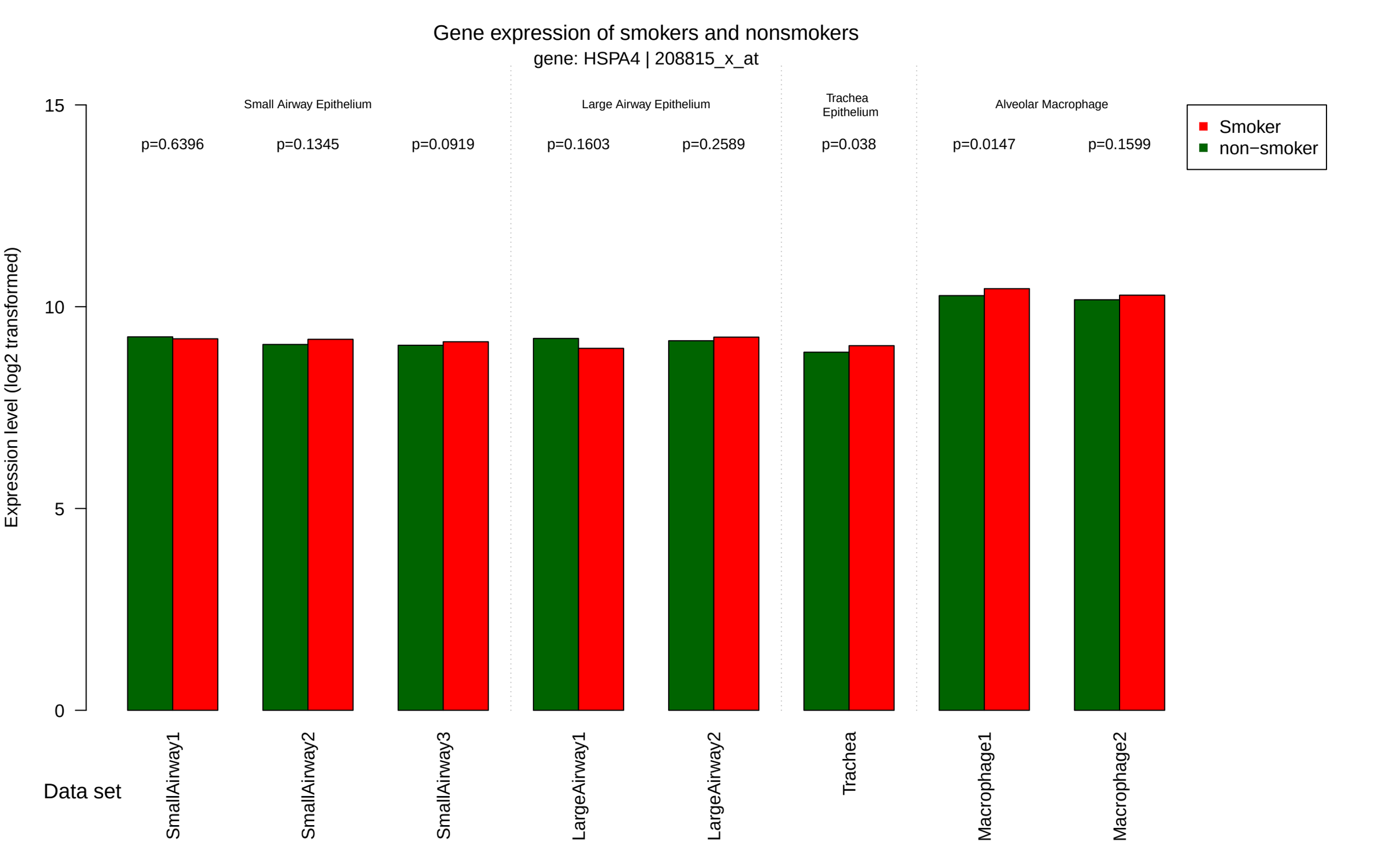
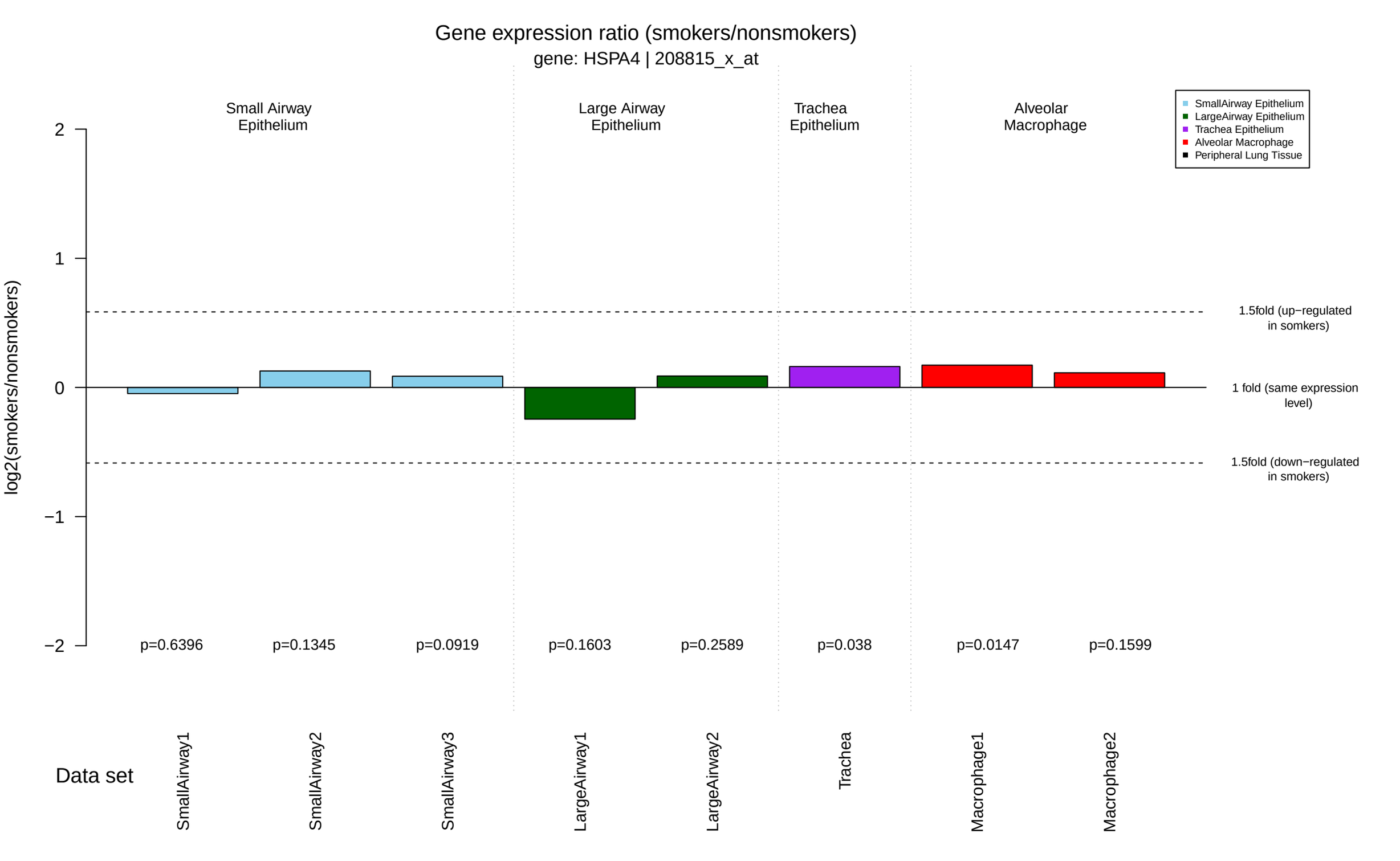
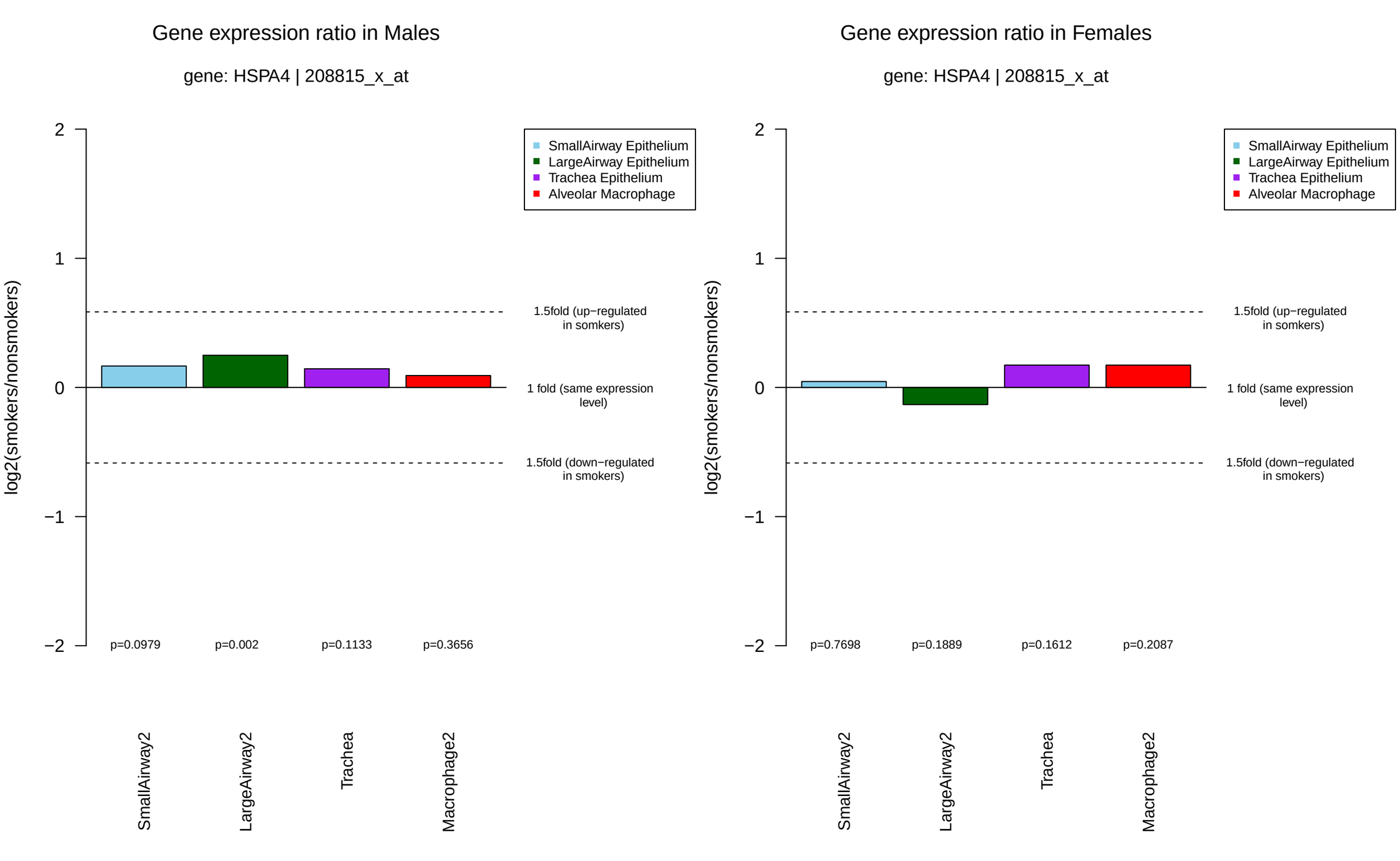
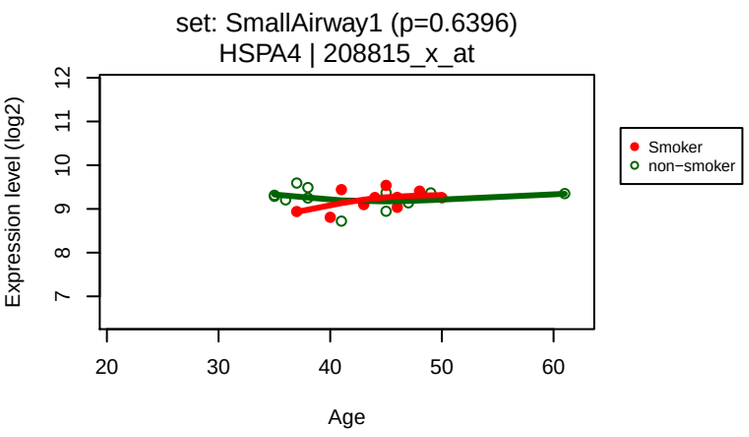
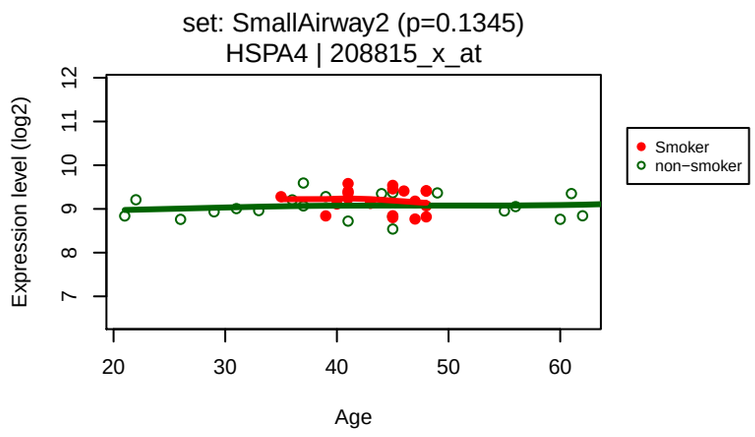
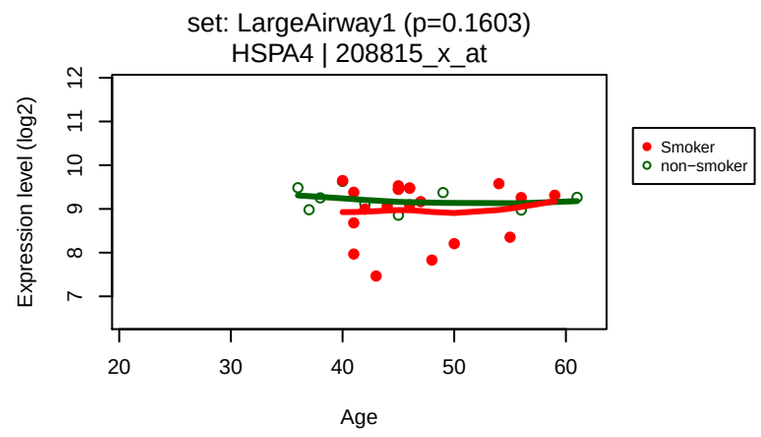

Gene expression of smokers and nonsmokers
gene: HSPA4 | 208815_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6396
p=0.1345
p=0.0919
p=0.1603
p=0.2589
p=0.038
p=0.0147
p=0.1599
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

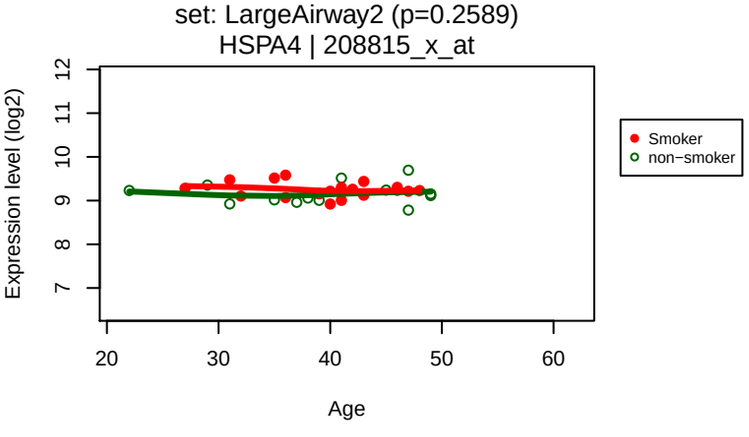

Gene expression ratio (smokers/nonsmokers)
gene: HSPA4 | 208815_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6396
p=0.1345
p=0.0919
p=0.1603
p=0.2589
p=0.038
p=0.0147
p=0.1599
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

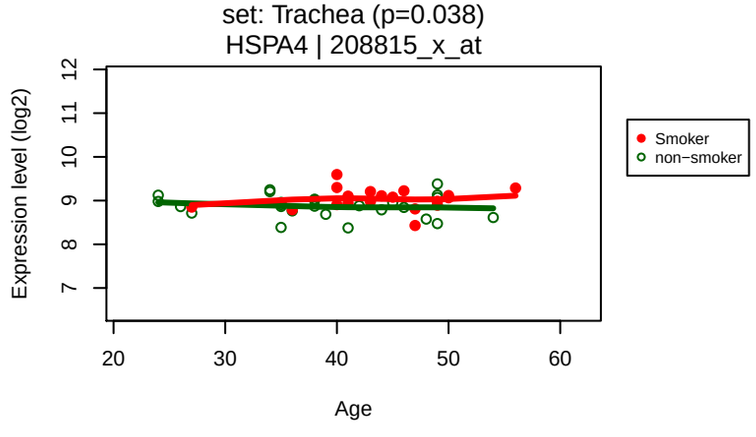

Gene expression ratio in Males
gene: HSPA4 | 208815_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0979
p=0.002
p=0.1133
p=0.3656
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA4 | 208815_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7698
p=0.1889
p=0.1612
p=0.2087
SmallAirway2
LargeAirway2
Trachea
Macrophage2

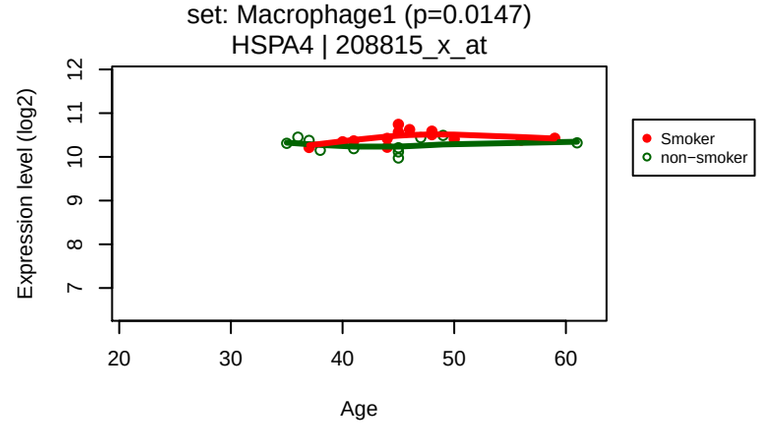

set: SmallAirway1 (p=0.6396)
HSPA4 | 208815_x_at
Expression level (log2)
Age
Smoker
non−smoker

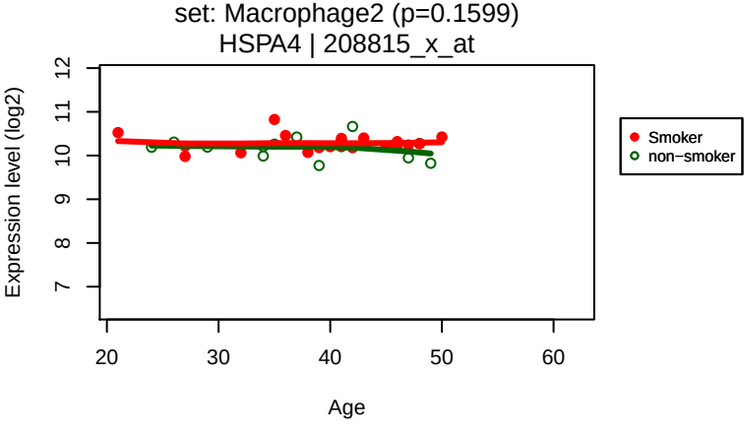

set: SmallAirway2 (p=0.1345)
HSPA4 | 208815_x_at
Expression level (log2)
Age
Smoker
non−smoker

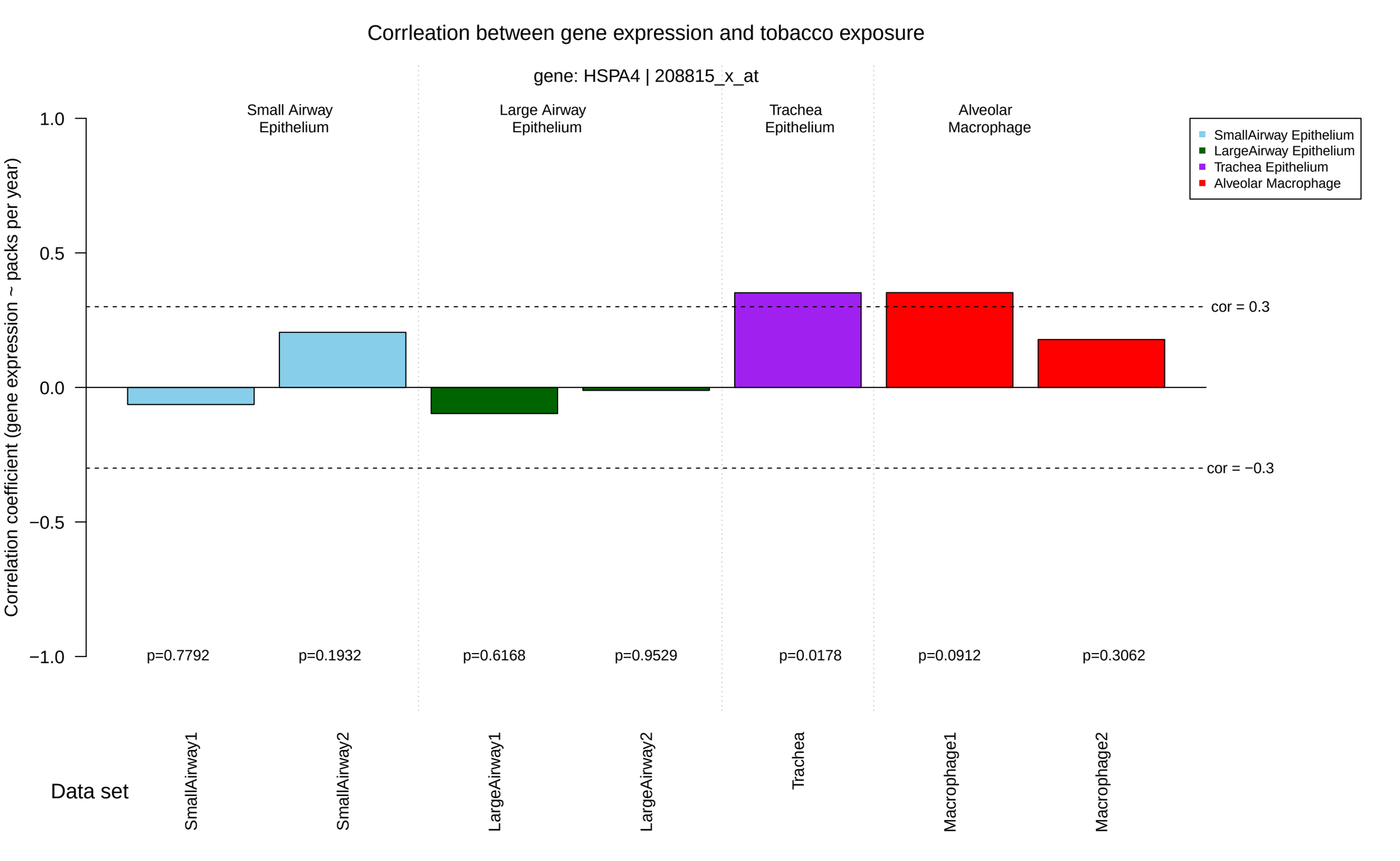

set: LargeAirway1 (p=0.1603)
HSPA4 | 208815_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2589)
HSPA4 | 208815_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.038)
HSPA4 | 208815_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0147)
HSPA4 | 208815_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1599)
HSPA4 | 208815_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA4 | 208815_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7792
p=0.1932
p=0.6168
p=0.9529
p=0.0178
p=0.0912
p=0.3062
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2